

Indian Institute of Technology Guwahati
Department of Electronics and Electrical Engineering
EE101 - Basic Electronics

Tutorial - 12

Date: Nov 16, 2023

1. For the network shown in Figure 1, Let $V_1 = 12V$, $V_2 = 10V$, $R_1 = 2k\Omega$, $R_2 = 1k\Omega$, $R_3 = 1k\Omega$, $R_4 = 1k\Omega$, $R_5 = 2k\Omega$, $R_6 = 1k\Omega$. Determine V_x and the voltage V_0

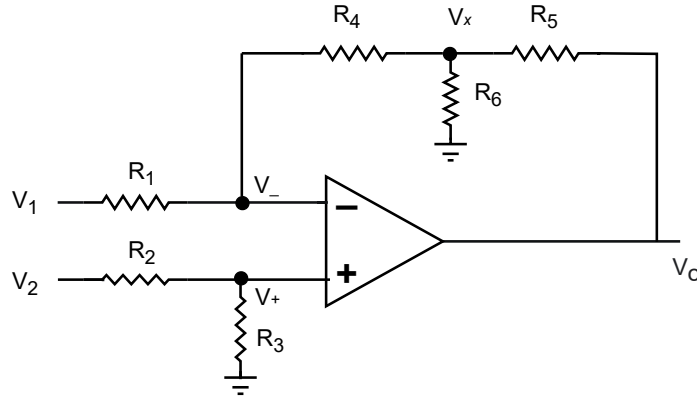


Figure 1

Solution

The voltage across R_3 is given as

$$\begin{aligned} V_+ &= \frac{V_2 R_3}{R_2 + R_3} \\ &= \frac{10V * 1k\Omega}{1k\Omega + 1k\Omega} \\ &= 5V \end{aligned}$$

As the OPAMP has input resistance of ∞ , $V_- = V_+ = 5V$

Also,

$$\begin{aligned} \frac{V_1 - V_-}{R_1} &= \frac{V_- - V_x}{R_4} \\ \frac{12 - 5}{2} &= \frac{5 - V_x}{1} \\ V_x &= 1.5V \end{aligned}$$

Similarly,

$$\begin{aligned} \frac{V_- - V_x}{R_4} &= \frac{V_x}{R_6} + \frac{V_x - V_0}{R_5} \\ \frac{5 - 1.5}{1} &= \frac{1.5}{1} + \frac{1.5 - V_0}{2} \\ V_0 &= -2.5V \end{aligned}$$

2. For the circuit shown in Fig. 2, Let $V_1 = 5V$, $V_2 = 4V$, $R_1 = 5k\Omega$, $R_2 = 5k\Omega$, $R_3 = 10k\Omega$, $R_4 = 10k\Omega$. Find the output voltage V_0

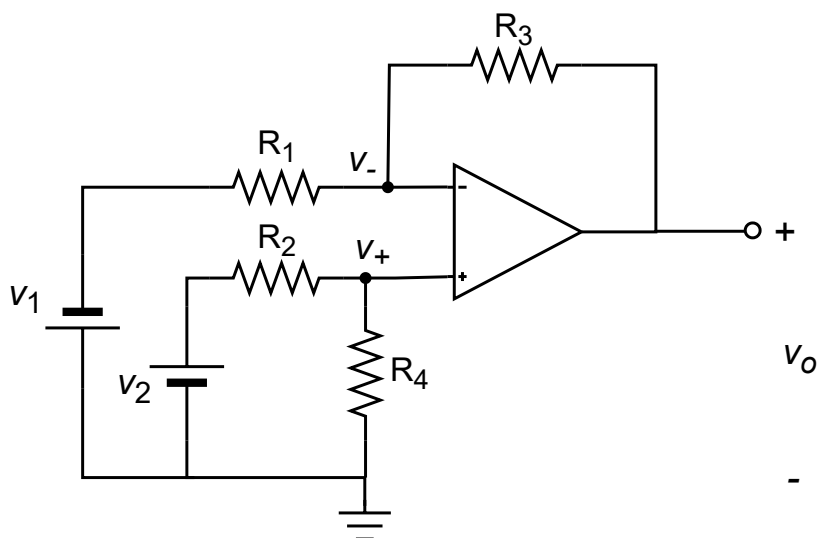


Figure 2

Solution

The Voltage across R_4 is given as

$$\begin{aligned} V_+ &= \frac{V_2 R_4}{R_2 + R_4} \\ &= \frac{4 * 10}{5 + 10} \\ &= 2.66V \end{aligned}$$

As the OPAMP has input resistance of ∞ , then $V_- = V_+ = 2.66V$

Also,

$$\begin{aligned} \frac{V_0 - V_-}{R_3} &= \frac{V_- - V_1}{R_1} \\ \frac{V_0 - 2.66}{10} &= \frac{2.66 - (-5)}{5} \\ V_0 &= 17.98V \end{aligned}$$

3. Calculate the mesh currents in the circuit below.

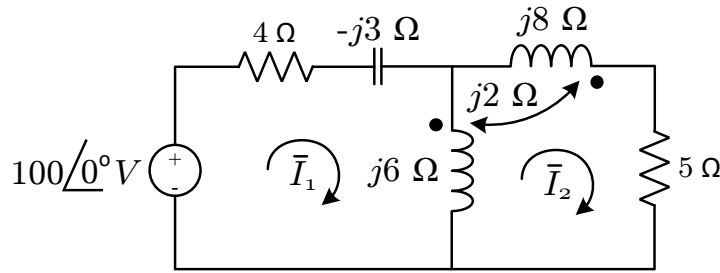


Figure 3

Solution:

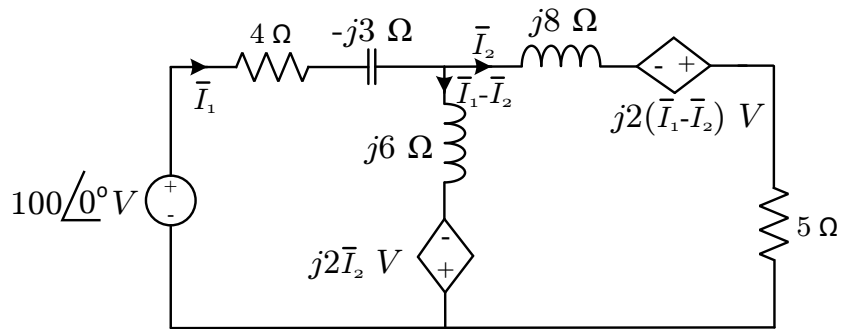


Figure 4

Applying KVL in mesh 1,

$$\begin{aligned} -100 + 4\bar{I}_1 - j3\bar{I}_1 + j6(\bar{I}_1 - \bar{I}_2) - j2\bar{I}_2 &= 0 \\ \Rightarrow (4 + j3)\bar{I}_1 - j8\bar{I}_2 &= 100 \end{aligned}$$

Applying KVL in mesh 2,

$$\begin{aligned} j8\bar{I}_2 + 5\bar{I}_2 + j6(\bar{I}_2 - \bar{I}_1) - j2(\bar{I}_1 - \bar{I}_2) + j2\bar{I}_2 &= 0 \\ \Rightarrow -j8\bar{I}_1 + (5 + j18)\bar{I}_2 &= 0 \end{aligned}$$

Putting the two mesh equations in matrix form,

$$\begin{bmatrix} 4 + j3 & -j8 \\ -j8 & 5 + j18 \end{bmatrix} \begin{bmatrix} \bar{I}_1 \\ \bar{I}_2 \end{bmatrix} = \begin{bmatrix} 100 \\ 0 \end{bmatrix}$$

The determinants are

$$\begin{aligned} \Delta &= \begin{vmatrix} 4 + j3 & -j8 \\ -j8 & 5 + j18 \end{vmatrix} = 30 + j87 \\ \Delta_1 &= \begin{vmatrix} 100 & -j8 \\ 0 & 5 + j18 \end{vmatrix} = 100(5 + j18) \\ \Delta_2 &= \begin{vmatrix} 4 + j3 & 100 \\ -j8 & 0 \end{vmatrix} = j800 \end{aligned}$$

So, the mesh currents are

$$\begin{aligned} \bar{I}_1 &= \frac{\Delta_1}{\Delta} = \frac{100(5 + j18)}{30 + j87} = \frac{1868.2/74.5^\circ}{92.03/71^\circ} = 20.3/3.5^\circ \text{ A} \\ \bar{I}_2 &= \frac{\Delta_2}{\Delta} = \frac{j800}{30 + j87} = \frac{800/90^\circ}{92.03/71^\circ} = 8.693/19^\circ \text{ A} \end{aligned}$$

4. Consider the circuit below. Determine the coupling coefficient. Calculate the energy stored in the coupled inductors at time $t = 1$ s if $v = 60 \cos(4t + 30^\circ)$ V. (60 is peak value of voltage)

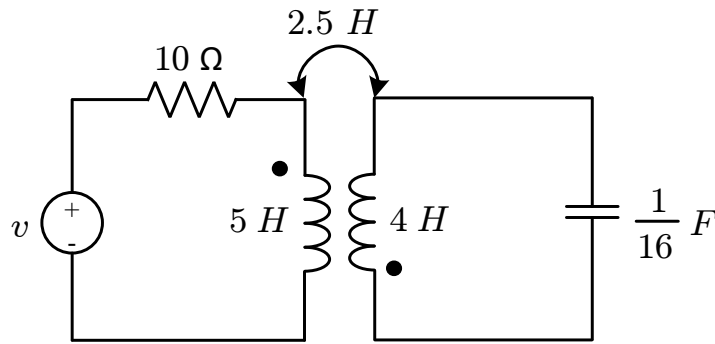


Figure 5

Solution:

The coupling coefficient is

$$k = \frac{M}{\sqrt{L_1 L_2}} = \frac{2.5}{\sqrt{20}} = 0.56$$

To find the energy stored, we need to calculate the current. To find the current, we need to obtain the frequency-domain equivalent of the circuit.

$$60 \cos(4t + 30^\circ) \Rightarrow 60\angle 30^\circ, \omega = 4 \text{ rad/s}$$

$$5 \text{ H} \Rightarrow j\omega L_1 = j20 \Omega$$

$$2.5 \text{ H} \Rightarrow j\omega M = j10 \Omega$$

$$4 \text{ H} \Rightarrow j\omega L_2 = j16 \Omega$$

$$\frac{1}{16} \text{ F} \Rightarrow \frac{1}{j\omega C} = -j4 \Omega$$

The frequency-domain equivalent circuit is shown below.

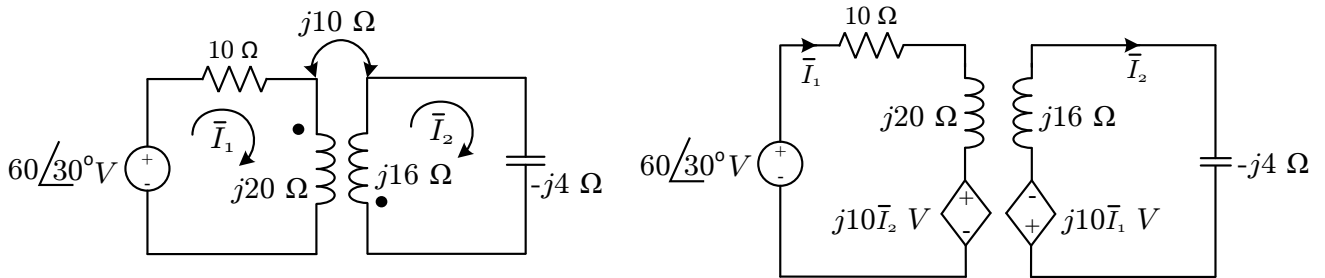


Figure 6

We now apply mesh analysis. For mesh 1,

$$(10 + j20)\bar{I}_1 + j10\bar{I}_2 = 60\angle 30^\circ$$

For mesh 2,

$$j10\bar{I}_1 + (j16 - j4)\bar{I}_2 = 0 \Rightarrow \bar{I}_1 = -1.2\bar{I}_2$$

Substituting this into the previous equation yields

$$\bar{I}_2(-12 - j14) = 60\angle 30^\circ \Rightarrow \bar{I}_2 = 3.254\angle 160.6^\circ \text{ A}$$

and

$$\bar{I}_1 = -1.2\bar{I}_2 = 3.905\angle -19.4^\circ \text{ A}$$

In the time domain,

$$i_1 = 3.905 \cos(4t - 19.4^\circ), \quad i_2 = 3.254 \cos(4t + 160.6^\circ)$$

At time $t = 1$ s, $4t = 4$ rad = 229.2°,

$$i_1 = 3.905 \cos(229.2^\circ - 19.4^\circ) = -3.389 \text{ A}$$

$$i_2 = 3.254 \cos(229.2^\circ + 160.6^\circ) = 2.824 \text{ A}$$

The total energy stored in the coupled inductor is

$$\begin{aligned} w(t) &= \frac{1}{2} L_1 i_1^2 + \frac{1}{2} L_2 i_2^2 + M i_1 i_2 \\ &= \frac{1}{2} (5) (-3.389)^2 + \frac{1}{2} (4) (2.824)^2 + 2.5 (-3.389) (2.824) = \mathbf{20.73 \text{ J}} \end{aligned}$$